Data-Backed Journal-Style Replication of *Patent Citations: An Analysis of Quality Differences and Citing Practices in Hybrid Corn*

Replication Paper Project

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Abstract

This paper presents a local calculated replication of Patent Citations: An Analysis of Quality Differences and Citing Practices in Hybrid Corn using only processed datasets and result files available in the workspace. The replication preserves the source paper's core empirical logic while making every local substitution explicit. The current local batch contains 712,375 observations across 345 entities, and the batch headline estimate is 0.0963 for Patent Quality with p = ¡ .001. The local analogue confirms that citation counts blend quality and citing practices: objective value and applicant-added citations are positive, examiner-added citations are negative, and other backward references remain strongly positive. The contribution is a journal-style replication manuscript that separates direct source claims from executable local evidence.

Keywords: patent citations; patent quality; applicant citations; examiner citations; local calculated analogue

1 Introduction

This replication begins from a paper-first reading of Patent Citations: An Analysis of Quality Differences and Citing Practices in Hybrid Corn and treats the source article as an empirical

argument that has to be rebuilt before it can be judged. The motivating question is: The source paper asks whether observed patent citations track objective quality or whether they also encode heterogeneous citing practices (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The study is worth revisiting because The replication matters because many strategy and innovation papers use citation counts as if they were clean measures of inventive quality. The original article does not rest on a descriptive regularity alone. It ties a theory about search, differentiation, timing, or organizational learning to a specific unit of analysis, sample frame, and estimand. A careful replication therefore has to recover the structure of that logic before any local coefficients are interpreted (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The executable evidence base used in this manuscript contains 712,375 observations across 345 entities and spans 2003 to 2018 in the local executable panel. Those numbers are not filler. They define the actual inferential scope of the local reconstruction and set the boundary on what can be claimed honestly from the available datasets (Moser et al., 2018; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper's contribution is a transparent measurement analogue that shows how much of the local citation record is attributable to quality versus channel-specific citing practices. The paper is written to read like a formal journal article rather than an internal workflow memo. The main text develops the theoretical logic, data reconstruction, derivations, empirical design, and results in continuous prose, while the appendices preserve the audit trail required for reruns and later extensions (Moser et al., 2018; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The paper belongs to a top-journal literature on patents as indicators, measurement error in innovation research, and the difficulty of separating knowledge value from citation behavior. (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Cohen and Levinthal, 1990; Kogut and Zander, 1992, 1996; Teece et al., 1997; Benner and Tushman, 2003). That literature matters because this replication asks whether the local citation record can be decomposed into an objective quality component and channel-specific citation practices using only verified processed files. (Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Henderson and Clark, 1990; Anderson and Tushman, 1986; Hargadon and Sutton, 1997; Tripsas and Gavetti, 2000; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Stuart, 2000). The present manuscript therefore treats the

source paper as one contribution inside a broader top-journal stream on exploration, recombination, performance measurement, alliance-based learning, inventive search, technological discontinuity, and patent-based inference. The literature review is intentionally broad because each replication paper should stand on a serious journal-level reference base rather than on the source article alone. (Rosenkopf and Almeida, 2003; Schilling and Phelps, 2007; Almeida and Kogut, 1999; Fleming et al., 2007; Jaffe, 1986; Jaffe et al., 1993; Trajtenberg, 1990; Hall et al., 2005; Henderson and Cockburn, 1994). Because the paper is fundamentally about measurement, the review also brings in classic work on patents as indicators, top-journal evidence on patent value, and adjacent strategy studies that rely on citations as dependent and independent variables. (Griliches, 1990; Jaffe, 1989). A second paper-specific literature block is added so that the manuscript's topic-matched references are carried into the final bibliography explicitly rather than being silently implied by the broader shared review. (Fleming and Sorenson, 2004; Nerkar, 2003).

2 Original Study, Theory, and Replication Boundary

The focal source article is (Moser et al., 2018). The original study can be summarized as a mechanism paper with a sharp empirical core. The original article evaluates patent citations as indicators rather than taking them as transparent measures of quality. Applicant and examiner citing practices are central to the argument. Objective patent quality is conceptually distinct from observed citations. That combination matters because the source article asks readers to believe not merely that two variables move together, but that a particular organizational or strategic mechanism explains why the relationship should appear in the data (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). In a setting like this, faithful replication requires more than reusing a title and rerunning a regression. The theoretical contribution lives in the way constructs are defined, how they enter the estimand, and which comparison the model is supposed to adjudicate. For that reason, this manuscript first reconstructs the published theory and only then turns to the local executable analogue (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The replication boundary is therefore explicit. Direct source facts are

carried over when they can be read from the published article itself. Local replication decisions appear only when the current workspace must translate the source design into verified processed datasets. The goal is disciplined mechanism recovery, not rhetorical imitation that quietly changes the design (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

2.1 Direct Source Facts Versus Replication Decisions

The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Citation practices are part of the empirical mechanism. The local reconstruction therefore adopts the following bounded decision: The local rerun uses applicant, examiner, and other backward citation counts from the cleaned workspace file. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that Objective quality is conceptually distinct from citations. The local reconstruction therefore adopts the following bounded decision: xi_real is used as the executable quality proxy. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). The manuscript distinguishes direct source facts from local execution decisions. The source article establishes that The paper is a patent-level design. The local reconstruction therefore adopts the following bounded decision: The sample is broadened from hybrid corn to the current processed patent panel while preserving the patent-level estimand. This bridge is treated as a visible design choice, not as a hidden substitution. The reason for spelling out this bridge in prose is that replication credibility depends on showing exactly where the manuscript is following the article and where it is making an explicit local translation. That is especially important in this

project, because the current paper is built from processed datasets already present in the workspace rather than from a reconstructed raw archival pipeline (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993).

DirectSourceFact	LocalReplicationDecision
Citation practices are part of the empirical mechanism.	The local rerun uses applicant, examiner, and other backward citation counts from the cleaned workspace file.
Objective quality is conceptually distinct from citations.	xi_real is used as the executable quality proxy.
The paper is a patent-level design.	The sample is broadened from hybrid corn to the current processed patent panel while preserving the patent-level estimand.

Direct source facts and explicit local execution decisions carried in the body of the paper.

3 Data, Sample, and Measures

The local evidence chain is deliberately modular. The source paper is treated as the authority for the research question, theoretical mechanism, and published design. Estimation, by contrast, proceeds only through the verified local files. The required dataset types in this paper are Processed DATASET2 patent-level quality panel with xi_real and forward citations. Cleaned backward citation file with applicant, examiner, and other-source categories. Local USPTO class and CPC complexity measures already materialized in the workspace (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005). Operationally, the reconstruction relies on four evidence layers: the analysis panel, the coefficient export, the sample summary, and, when available, the fit summary. The resulting executable panel supplies 712,375 observations across 345 entities and covers 2003 to 2018 in the local executable panel. Every numeric claim in the results section is intended to trace back to

one of those verified evidence layers rather than to a hand-copied number from the article (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This data-lineage discipline matters for two reasons. First, it prevents the manuscript from smuggling unverified preprocessing steps into the design. Second, it forces every substitution to be visible. If a construct cannot be reproduced from the current processed datasets, the paper states the analogue directly instead of pretending that the original archive has been fully recreated (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The local analogue combines the processed DATASET2 quality panel with applicant, examiner, and other backward-citation channels from the cleaned local citation pipeline. The local sample frame is Processed utility-patent panel with objective quality and cleaned backward citation channels The observed time window in the executable panel is 2003 to 2018 in the local executable panel The outcome is Log forward citations per patent (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005). The main text works from a verified analysis panel, a verified coefficient export, and a verified sample summary. Exact file names are retained in the replication notes and appendices, while the article body focuses on data lineage and empirical function (Moser et al., 2018; Hall et al., 2005).

3.1 Executable File Selection and Data Availability

The present manuscript works from the processed datasets and calculated exports already available in the workspace for Moser, Ohmstedt, and Rhode (2018). That does not mean the paper skips data analysis. It means the data analysis begins from the current executable panel, current coefficient export, and current sample-summary files rather than from a freshly rebuilt raw archive. The substantive obligation is therefore to explain exactly why each verified file is needed and how it enters the paper's argument (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005). The analysis panel used in this paper serves a specific empirical role: it is needed to estimate the local model and regenerate descriptive statistics. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The coefficient export used

in this paper serves a specific empirical role: it is needed to recover the reported local coefficients and p-values. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The sample summary used in this paper serves a specific empirical role: it is needed to audit sample construction and raw grouped moments. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The fit summary used in this paper serves a specific empirical role: it is needed to check model diagnostics when a fit file is available. This evidence layer is not listed merely for bookkeeping. It is the concrete object that allows the manuscript to calculate the sample moments, estimators, and interpretation discussed in the main text. The body of the paper keeps the emphasis on evidence and function rather than on machine-specific file labels (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, these files form the minimum executable evidence chain behind the paper. The theory comes from the source article, the estimable sample comes from the processed analysis panel, the regression narrative comes from the current coefficient export, and the sample audit comes from the local summary file. That is the basis on which the manuscript makes every empirical claim (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

Stage	NeededFor	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	Yes	Executable local dataset

Stage	NeededFor	Exists	Status
Coefficient export	Recover the reported local coefficients and p-values	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	Yes	Executable local result file
Fit summary	Check model diagnostics when a fit file is available	Yes	Executable local result file

Verified evidence layers used to support the paper's calculations.

3.2 Sample Reconstruction

The local analysis file contains 712,375 observations drawn from 345 entities. The observed time span runs from 2003 to 2018. Selected sample moments are Xi Real = 18.29, Cites = 8.860, Backward Applicant = 6.475, Backward Examiner = 2.132, Backward Other = 0.949 (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005). These sample moments matter because they anchor the interpretation of the replication design. They indicate not only the scale of the local panel but also the baseline level of the key outcome and explanatory variables before the formal estimating equation is imposed (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

Statistic	Value
Observations	712,375
Unique Firms	345
Start Year	2003
End Year	2018
Mean Xi Real	18.29
Mean Cites	8.860

Statistic	Value
Mean Backward Applicant	6.475
Mean Backward Examiner	2.132
Mean Backward Other	0.949

Sample reconstruction from the current local dataset.

3.3 Variable Construction and Derivation

The construct Objective quality is operationalized locally as xi_real and appears in the executable dataset as xi_real. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Carry forward the processed objective-value variable from DATASET2.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Observed patent citations is operationalized locally as Forward citation count and appears in the executable dataset as cites; log_cites. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use local observed forward citation counts and log-transform them.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja,

2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Applicant citing practices is operationalized locally as Applicant-added backward citations and appears in the executable dataset as backward_applicant; log_backward_applicant. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Count applicant-labeled backward citations from the cleaned citation file.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Examiner citing practices is operationalized locally as Examiner-added backward citations and appears in the executable dataset as backward_examiner; log_backward_examiner. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Count examiner-labeled backward citations from the same file.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005). The construct Technology complexity is operationalized locally as CPC group count and appears in the executable dataset as cpc_group_count; log_cpc_group_count. The derivation is not inferred loosely from the source article; it follows the verified local pipeline: Use the processed CPC breadth count as a technology-complexity control.. This mapping is retained because it preserves the construct's functional role in the original empirical argument while keeping the measurement auditable from the processed files used in estimation. The practical importance of this choice is that the manuscript can show

how the concept enters the current dataset, how it is scaled, and how it interacts with the rest of the specification instead of treating the variable as a black box. In replication terms, this is the point where conceptual comparability and executable traceability have to be balanced carefully, and the chosen local definition is intended to preserve that balance rather than maximize superficial similarity (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

Construct	Local measure	Derivation
Objective quality	xi_real	Carry forward the processed objective-value variable from DATASET2.
Observed patent citations	Forward citation count	Use local observed forward citation counts and log-transform them.
Applicant citing practices	Applicant-added backward citations	Count applicant-labeled backward citations from the cleaned citation file.
Examiner citing practices	Examiner-added backward citations	Count examiner-labeled backward citations from the same file.
Technology complexity	CPC group count	Use the processed CPC breadth count as a technology-complexity control.

Variable mapping and derivation logic in the main text.

3.4 Construct-Level Derivation Logic

The derivation appendix records how each major construct enters the executable estimand for Moser, Ohmstedt, and Rhode (2018). The point is not to rehearse notation for its own sake, but to show how the source paper's conceptual vocabulary is translated into verified columns in the local panel. For Objective quality, the manuscript uses xi_real under the exact local implementation xi_real. The derivation step is: Carry forward the processed objective-value variable from DATASET2.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Observed patent citations, the manuscript uses Forward citation count under the exact local implementation cites;

log_cites. The derivation step is: Use local observed forward citation counts and log-transform them.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Applicant citing practices, the manuscript uses Applicant-added backward citations under the exact local implementation backward_applicant; log_backward_applicant. The derivation step is: Count applicant-labeled backward citations from the cleaned citation file.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Examiner citing practices, the manuscript uses Examiner-added backward citations under the exact local implementation backward_examiner; log_backward_examiner. The derivation step is: Count examiner-labeled backward citations from the same file.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept. For Technology complexity, the manuscript uses CPC group count under the exact local implementation cpc_group_count; log_cpc_group_count. The derivation step is: Use the processed CPC breadth count as a technology-complexity control.. This construct is retained because it occupies the same analytic position in the local design that the published paper assigns to the original concept.

4 Descriptive Analysis of the Current Dataset

The current dataset yields a richer descriptive picture than a single mean or coefficient can convey. Patent quality has mean 18.29, standard deviation 35.09, and an interquartile range from 3.688 to 19.06; Forward citations has mean 8.860, standard deviation 51.01, and an interquartile range from 0.00000 to 5.000; Log forward citations has mean 1.135, standard deviation 1.188, and an interquartile range from 0.00000 to 1.792; Applicant backward citations has mean 6.475, standard deviation 56.62, and an interquartile range from 0.00000 to 1.000 (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The time-series view is equally informative. Patents changes from 33,777 in the first observed period to 57,171 in the last observed period; MeanQuality changes from 20.10 in the first observed period to 29.66 in

the last observed period; MeanCites changes from 19.85 in the first observed period to 1.642 in the last observed period . These temporal shifts help explain why a paper-comparable replication has to describe the support of the data before turning to regression output (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The regressor-quantile analysis shows that mean outcomes range from 0.911 to 1.230 across the support of the focal regressor, while the mean regressor itself ranges from 0.956 to 3.905. That pattern is useful because it shows what the formal model is trying to account for in the current dataset (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The correlation structure is reported in a compact form because it helps distinguish whether the main explanatory variables move in lockstep with the controls or whether the regression is drawing on independent variation. That is especially important in these replications, where construct translation can otherwise make the specification look more deterministic than it really is (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

4.1 Data Quality, Coverage, and Support

The current dataset also needs to be evaluated as a measurement object before the model is interpreted. CPC group count has missing share 0.19%; Log CPC group count has missing share 0.19%; Patent quality has missing share 0.00% . At the other end of the support, Patent quality remains close to complete with missing share 0.00%; Forward citations remains close to complete with missing share 0.00%; Log forward citations remains close to complete with missing share 0.00% . This matters because the credibility of the regression depends on whether the focal constructs survive sample cleaning with enough support to be interpreted substantively (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005). Coverage across entities is also visible in the current panel rather than assumed. The five most heavily observed entities contribute between 25,638 and 99,013 observations each. The purpose of reporting this concentration pattern is to show whether the estimated mechanism is being learned from a broad panel or from a narrow set of heavily observed entities

(Moser et al., 2018; Jaffe, 1986; Hall et al., 2005). The support of the focal regressor is broad enough to permit a distributional reading in addition to the formal model. In the lowest observed quantile, the mean regressor is 0.956 and the mean outcome is 0.911 across 142,511 observations. In the highest observed quantile, the corresponding values are 3.905, 1.182, and 142,475 observations. This shift matters because it shows that the regression is being estimated on economically different parts of the observed support rather than on nearly identical cells (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005). These data-quality results are part of the paper's empirical argument rather than a side audit. The replication matters because many strategy and innovation papers use citation counts as if they were clean measures of inventive quality. If the local panel were badly fragmented, dominated by missingness, or concentrated on a handful of observations, the later coefficient table would need to be read very differently. Bringing this evidence into the body of the article is therefore necessary for an honest replication (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005).

Variable	Missing	MissingShare
Patent quality	0	0.00%
Forward citations	0	0.00%
Log forward citations	0	0.00%
Applicant backward citations	0	0.00%
Examiner backward citations	0	0.00%
Other backward citations	0	0.00%
CPC group count	1,360	0.19%
Log patent quality	0	0.00%
Log applicant citations	0	0.00%
Log examiner citations	0	0.00%
Log other citations	0	0.00%
Log CPC group count	1,360	0.19%

Missingness profile for variables analyzed in the body of the paper.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Patent quality	712,375	18.29	35.09	3.688	8.255	19.06	0	2718
Forward citations	712,375	8.860	51.01	0.0000	1.000	5.000	0	3222
Log forward citations	712,375	1.135	1.188	0.0000	0.693	1.792	0	8
Applicant backward citations	712,375	6.475	56.62	0.0000	0.0000	1.000	0	2863
Examiner backward citations	712,375	2.132	4.753	0.0000	0.0000	3.000	0	292
Other backward citations	712,375	0.949	9.696	0.0000	0.0000	0.0000	0	1014
CPC group count	711,015	7.350	7.591	3.000	5.000	9.000	1	293
Log patent quality	712,375	2.322	1.051	1.545	2.225	2.999	0	8
Log applicant citations	712,375	0.570	1.085	0.0000	0.0000	0.693	0	8
Log examiner citations	712,375	0.629	0.884	0.0000	0.0000	1.386	0	6

Main descriptive statistics discussed in the body of the paper.

Period	Patents	MeanQuality	MeanCites	MeanApplicant	MeanExaminer
2003	33,777	20.10	19.85	0.149	1
2004	32,280	19.40	17.91	0.320	1
2005	34,501	18.93	16.34	0.486	1
2006	34,295	17.26	14.84	0.875	2
2007	37,364	15.33	12.72	1.786	2
2008	40,514	13.03	10.93	2.568	2
2009	34,006	13.72	10.03	3.431	2
2010	36,300	14.36	10.59	5.029	2
2011	41,541	14.73	9.900	6.289	2
2012	51,446	14.73	7.906	7.836	2
2013	52,128	15.59	6.851	8.056	2
2014	52,113	17.01	6.474	10.28	2
2015	56,363	17.78	5.459	10.70	3
2016	58,435	20.57	4.619	11.92	3
2017	60,141	23.63	2.705	10.71	3

Period	Patents	MeanQuality	MeanCites	MeanApplicant	MeanExaminer
2018	57,171	29.66	1.642	9.028	2

Observed time patterns in the current dataset.

Quantile	N	MeanRegressor	MeanOutcome
Q1	142,511	0.956	0.911
Q2	142,442	1.693	1.144
Q3	142,476	2.228	1.210
Q4	142,471	2.825	1.230
Q5	142,475	3.905	1.182

Grouped summaries used to describe raw patterns in the current dataset.

Variable	Patent quality	Forward citations	Log forward citations	Applicant backward citations	Examiner backward citations	Other backward citations	CPC group count	Log patent quality	Log applicant citations	Log examiner citations	Log other citations	Log CPC group count
Patent quality	1.000	0.028	0.010	0.037	0	0.004	0	0.704	0.004	0	0.004	0
Forward citations	0.028	1.000	0.460	0.294	0	0.133	0	0.063	0.090	0	0.095	0
Log forward citations	0.010	0.460	1.000	0.123	0	0.110	0	0.067	-0.065	0	0.192	0
Applicant backward citations	0.037	0.294	0.123	1.000	0	-0.011	0	0.067	0.467	0	-0.030	0
Examiner backward citations	-0.034	-0.002	-0.048	0.086	1	0.017	0	-0.075	0.303	1	0.030	0
Other backward citations	0.004	0.133	0.110	-0.011	0	1.000	0	0.017	-0.051	0	0.603	0

Compact correlation structure for the main variables.

5 Empirical Strategy and Identification

The estimand targeted in this manuscript is A patent-level fixed-effects regression of log forward citations on log objective quality and different backward-citation channels. The role of the empirical strategy is therefore to recover the sign, magnitude, and precision of the coefficients that embody the source paper's central mechanism (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993). Identification remains observational. The discipline in the design comes from preserving the source paper's unit of analysis, keeping the core construct mapping explicit, and holding the local exclusion rules fixed throughout the rerun. This does not convert the study into an experimental design, but it does make the inferential claim narrower and more honest (Moser et al., 2018; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The first set of replication decisions concerns scope. The unit of analysis is Patent, the sample frame is Processed utility-patent panel with objective quality and cleaned backward citation channels, and the empirical window is 2003 to 2018 in the local executable panel. These choices define what counts as a comparable observation in the local reconstruction and prevent the paper from drifting into a broader but theoretically mismatched sample (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The second set of decisions concerns measurement and specification. The outcome is Log forward citations per patent, and the main explanatory design is Fixed-effects regression of observed citations on objective quality and applicant, examiner, and other backward citation channels. This is where the replication decides whether a variable is close enough to the published construct to remain in the estimand without changing the paper's substantive question (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The third set of decisions concerns statistical discipline. Controls and fixed effects are handled as follows: Filing-year and primary-class fixed effects plus local technology-complexity control.. The crucial exclusion or substitution rule is: The local rerun keeps utility patents with verified quality and class information and uses xi_real as the executable quality proxy.. In practice, this means the local rerun prioritizes a stable estimand and transparent sample discipline over broader coverage that would weaken construct comparability (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). A final design choice concerns interpretation. The manuscript treats the replication as successful when

the local evidence can transparently evaluate the same mechanism the source paper emphasized. When the local data architecture differs, the paper marks the result as a calculated analogue rather than claiming byte-identical duplication (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

DecisionPoint	LocalSpecification
Research question	The source paper asks whether observed patent citations track objective quality or whether they also encode heterogeneous citing practices.
Unit of analysis	Patent
Sample frame	Processed utility-patent panel with objective quality and cleaned backward citation channels.
Time window	2003 to 2018 in the local executable panel.
Outcome definition	Log forward citations per patent.
Main explanatory design	Fixed-effects regression of observed citations on objective quality and applicant, examiner, and other backward citation channels.
Controls and fixed effects	Filing-year and primary-class fixed effects plus local technology-complexity control.
Exclusion or substitution logic	The local rerun keeps utility patents with verified quality and class information and uses xi_real as the executable quality proxy.

Replication decision register retained in the main text.

5.1 Why These Design Decisions Are Necessary

Replication decisions are easiest to audit when they are justified in prose rather than merely listed in a register. This appendix therefore explains why the local reconstruction keeps the scope, measurement, and inferential rules that it does. Scope decisions: the manuscript works at the level of Patent inside the sample frame Processed utility-patent panel with objective quality and cleaned backward citation channels over 2003 to 2018 in the local executable panel. These are not cosmetic constraints. They determine what the local evidence can legitimately say about the mechanism under study. Measurement decisions: the outcome is Log forward citations per patent and the main explanatory design is Fixed-effects regression of observed citations on objective quality and applicant, examiner, and other backward citation channels. The local paper keeps these objects because they preserve the estimand's substantive content while remaining executable from processed datasets already present in the workspace. Inference decisions: Filing-year and primary-class fixed effects plus local technology-complexity control. The main exclusion or substitution rule is The local rerun keeps utility patents with verified quality and class information and uses xi_real as the executable quality proxy. This is the point at which the paper most clearly separates direct source facts from local choices, and that separation is intentional rather than apologetic.

$$\log(1 + \text{Cites}_p) = \beta_1 \log(1 + \text{Quality}_p) + \beta_2 \log(1 + \text{ApplicantBack}_p) + \beta_3 \log(1 + \text{ExaminerBack}_p) + \beta_4 \log(1 + \text{Ot} \tag{1}$$

6 Results

6.1 Main Estimates

The central local estimate is 0.0963 on Patent Quality (p = ¡ .001). The local analogue confirms that citation counts blend quality and citing practices: objective value and applicant-added citations are positive, examiner-added citations are negative, and other backward references remain strongly positive. Other backward citations is estimated at 0.171 with p = ¡ .001. Technology complexity is estimated at 0.202 with p = ¡ .001. Applicant backward citations is estimated at 0.0779 with p = ¡ .001 (Moser et al., 2018; Hall et al., 2005). The coefficient table below reports the compact pattern

most relevant to the source paper's central claim. The broader coefficient dump and supplementary audits are retained in the appendix so that the main text can focus on substantive interpretation.

Term	Estimate	Std. error	P-value
Patent quality	0.0963	0.0167	1.47e-08
Applicant backward citations	0.0779	0.0103	2.55e-13
Examiner backward citations	-0.0491	0.0114	2.05e-05
Other backward citations	0.1713	0.0134	6.43e-32
Technology complexity	0.2018	0.0175	7.29e-27

Key coefficient table used in the main text.

6.2 Detailed Interpretation of the Local Results

The coefficient pattern is not being summarized as a bare sign check. Other backward citations is estimated at 0.171 with a p-value of ¡ .001; Technology complexity is estimated at 0.202 with a p-value of ¡ .001; Applicant backward citations is estimated at 0.0779 with a p-value of ¡ .001; Patent quality is estimated at 0.0963 with a p-value of ¡ .001 (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Within the focal specification, the sharpest local signal comes from Other backward citations, which is estimated at 0.171 with p = ¡ .001. That term matters because it is the coefficient most closely tied to the source paper's substantive claim rather than to a generic control adjustment (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The scale of the underlying variables also matters for interpretation. The outcome Log forward citations has mean 1.135 and standard deviation 1.188 in the local panel, while the main regressor Log patent quality has mean 2.322 and standard deviation 1.051 (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). A simple grouped comparison also helps clarify the pattern. Across the regressor quantiles, the mean outcome ranges from 0.911

to 1.230 while the mean regressor ranges from 0.956 to 3.905. This does not substitute for the model, but it shows how much raw variation the specification must explain (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). Taken together, the descriptive evidence and formal estimates should be read as a structured assessment of the paper's mechanism. The local analogue confirms that citation counts blend quality and citing practices: objective value and applicant-added citations are positive, examiner-added citations are negative, and other backward references remain strongly positive. The substantive question is not whether every coefficient matches a published table entry digit for digit, but whether the local evidence supports the same directional and conceptual interpretation (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005).

6.3 Coefficient Families in the Current Estimation Output

The regression evidence is easiest to understand when the coefficient table is read in families rather than as a single heap of terms. The paper therefore separates the estimates by model and highlights the strongest positive, strongest negative, and most focal signals within each specification (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). Within baseline, the leading coefficient family reads as follows: Other backward citations = 0.171 (p = ¡ .001); Technology complexity = 0.202 (p = ¡ .001); Applicant backward citations = 0.0779 (p = ¡ .001); Patent quality = 0.0963 (p = ¡ .001); Examiner backward citations = -0.0491 (p = ¡ .001). The largest positive estimate in this specification is Technology complexity at 0.202, while the largest negative estimate is Examiner backward citations at -0.0491. Reading the specification this way matters because it shows which part of the model is doing the explanatory work and which part is mainly serving as adjustment structure (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). This fuller coefficient reading is especially important for Moser, Ohmstedt, and Rhode (2018) because the source paper's contribution is not exhausted by one headline estimate. The current dataset can confirm some parts of the published mechanism more strongly than

others, and a formal replication paper should show that uneven pattern directly in the main text (Moser et al., 2018; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996).

6.4 Model Comparison Using the Current Dataset

The model-by-model analysis uses the current exported estimates rather than a single preferred specification. In Baseline, the most informative focal term is Other backward citations with estimate 0.171 and p = ¡ .001 (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). The Baseline specification contributes its own reading to the replication. Other backward citations is estimated at 0.171 with p = ¡ .001 and is classified as a strong signal; Technology complexity is estimated at 0.202 with p = ¡ .001 and is classified as a strong signal; Applicant backward citations is estimated at 0.0779 with p = ¡ .001 and is classified as a strong signal . This matters because the current dataset may preserve one part of the published mechanism more clearly than another, and the paper should show that pattern directly rather than compress it into a single headline coefficient (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This model comparison matters for Moser, Ohmstedt, and Rhode (2018) because the source article's claim can only be evaluated responsibly if the pattern persists across the outcome or specification variants that are actually present in the local results folder. Put differently, the current dataset should be allowed to show where the mechanism is stable, where it weakens, and where it changes sign under a different outcome definition (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

Term	Model	Estimate	PValue	Signal	Reading
Patent quality	Baseline	0.0963	¡ .001	Strong	Positive estimate with p = ¡ .001.
Applicant backward citations	Baseline	0.0779	¡ .001	Strong	Positive estimate with p = ¡ .001.

Term	Model	Estimate	PValue	Signal	Reading
Examiner backward citations	Baseline	-0.0491	¡ .001	Strong	Negative estimate with p = ¡ .001.
Other backward citations	Baseline	0.1713	¡ .001	Strong	Positive estimate with p = ¡ .001.
Technology complexity	Baseline	0.2018	¡ .001	Strong	Positive estimate with p = ¡ .001.

Model-by-model interpretation table drawn from the current coefficient exports.

6.5 Diagnostic Evidence

Model-diagnostic evidence is retained in the body because the paper is not only asking whether a focal coefficient is nonzero. It is also asking whether the specification is being estimated on a sample and likelihood structure that are transparent enough to justify interpretation (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005). For citation_quality, the current fit export reports N = 710,951, pseudo R-squared = 0.0896 (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005). These diagnostics do not, by themselves, validate the source paper's theory. They do, however, show that the local data analysis behind Moser, Ohmstedt, and Rhode (2018) is being carried out on a clearly described estimating sample with observable fit statistics (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005).

model	n	logLik	AIC	BIC	pseudo_r2
citation_quality	710,951	-1,029,755	2,060,411	2,065,586	0.0896

Model fit statistics discussed in the main results section.

Term	Estimate	Std. error	Statistic	P-value
Patent quality	0.0963	0.0167	5.776	1.47e-08
Applicant backward citations	0.0779	0.0103	7.554	2.55e-13

Term	Estimate	Std. error	Statistic	P-value
Examiner backward citations	-0.0491	0.0114	-4.308	2.05e-05
Other backward citations	0.1713	0.0134	12.78	6.43e-32
Technology complexity	0.2018	0.0175	11.50	7.29e-27

Detailed coefficient table from the current local estimation output.

7 Comparison to the Published Study

7.1 Original Results Versus the Current Local Dataset

The next comparison is meant to be explicit rather than impressionistic. The original paper's central empirical claim, the main measurement layer, and the effective sample definition are set alongside the current local dataset result so the reader can see where agreement is substantive and where it is conditional on different data architecture. (Moser et al., 2018). For Moser, Ohmstedt, and Rhode (2018), the local headline result is Patent Quality = 0.0963 with p = ¡ .001. The value of presenting that estimate directly against the original paper's reported finding is that it shows whether the mechanism survives a different but transparent processed dataset rather than asking the reader to infer comparability from prose alone.

Aspect	OriginalPaper	LocalDataset	Reading
Main empirical result	The published paper argues that patent citations partly reflect quality but also incorporate systematic citing practices, with nonpatent references offering a cleaner signal of public-research knowledge flows.	Patent Quality = 0.0963 with p = ¡ .001.	The local evidence reinforces the central claim that citations are not pure quality signals. Objective patent value matters, but the citation record also reflects applicant, examiner, and other-source citing practices.
Measurement and data construction	The published design separates objective quality differences from applicant and examiner citing behavior in a historically specific patent setting.	The local analogue combines the processed DATASET2 quality panel with applicant, examiner, and other backward-citation channels from the cleaned local citation pipeline.	This row isolates how the executable local dataset differs from the original measurement architecture while preserving the same substantive target.
Sample and unit of analysis	The published sample is a patent-level study of hybrid corn patents and their citation practices.	Processed utility-patent panel with objective quality and cleaned backward citation channels. Local sample = 712,375 observations across 345 entities.	This row clarifies whether agreement or disagreement with the original paper could reflect sample translation rather than theory alone.

Comparison of the original paper's results and design to the current local dataset analysis.

7.2 Interpretive Implications of the Comparison

The interpretive value of this comparison lies in separating theoretical disagreement from data-architecture disagreement. The published study relies on the following measurement layer: The published design separates objective quality differences from applicant and examiner citing behavior in a historically specific patent setting By contrast, the local paper relies on the following executable design: The local analogue combines the processed DATASET2 quality panel with applicant, examiner, and other backward-citation channels from the cleaned local citation pipeline These are not trivial implementation details. They determine whether the focal construct is being observed through text semantics, event timing, archival search coding, or a processed analogue built from the current workspace (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The same caution applies to the sample itself. The published design is described as follows: The published sample is a patent-level study of hybrid corn patents and their citation practices The local sample instead works from Processed utility-patent panel with objective quality and cleaned backward citation channels and yields 712,375 observations across 345 entities. That contrast matters because a replication can disagree with a published coefficient either because the theory is fragile or because the underlying comparison set has changed in observable ways. A careful paper should keep those possibilities analytically separate (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). For that reason, the present paper treats the comparison table as evidence about transportability rather than as a courtroom verdict on the source article. The local evidence reinforces the central claim that citations are not pure quality signals. Objective patent value matters, but the citation record also reflects applicant, examiner, and other-source citing practices. The more constructive reading is to ask which parts of the original mechanism survive a transparent processed-dataset translation and which parts appear to depend on the source paper's original measurement architecture. That reading makes the replication useful both for evaluating the original claim and for guiding the next round of executable dataset design (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and

Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The comparison to the published paper is intentionally narrow. The local evidence reinforces the central claim that citations are not pure quality signals. Objective patent value matters, but the citation record also reflects applicant, examiner, and other-source citing practices. The manuscript therefore evaluates concordance at the level of mechanism, sign pattern, and construct behavior before drawing any conclusion about exact numerical replication (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). This comparison rule matters because the local paper is based only on processed datasets and verified outputs already present in the workspace. When the original paper depends on proprietary, historical, or otherwise unavailable preprocessing layers, a full byte-identical recreation is not claimed. The more defensible question is whether the local evidence would lead a careful reader to a similar substantive interpretation (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001).

8 Discussion

The local evidence reinforces the central claim that citations are not pure quality signals. Objective patent value matters, but the citation record also reflects applicant, examiner, and other-source citing practices (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001). Taken together, the results are most informative about whether the source paper's mechanism is directionally recoverable in local data rather than whether every original coefficient can be duplicated exactly (Moser et al., 2018; Jaffe, 1986; Hall et al., 2005; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996). The first limitation concerns measurement. The source article studies hybrid corn and a source-specific historical setting (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The second limitation concerns sample translation. The local analogue instead uses the broad processed patent panel currently available in the workspace (Moser et al., 2018; March, 1991; Levitt and March,

1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005). The third limitation concerns interpretation. The paper therefore reproduces the measurement logic rather than the original product-market archive (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Jaffe, 1986; Hall et al., 2005).

9 Conclusion

The paper's contribution is a transparent measurement analogue that shows how much of the local citation record is attributable to quality versus channel-specific citing practices (Moser et al., 2018; March, 1991; Levitt and March, 1988; Levinthal and March, 1993; Fleming, 2001; Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Rosenkopf and Nerkar, 2001; Jaffe, 1986; Hall et al., 2005). The paper therefore contributes a traceable local replication record that can be rerun, audited, and extended as additional datasets or original-source details become available (Moser et al., 2018; Ahuja, 2000; Ahuja and Katila, 2001; Stuart and Podolny, 1996; Jaffe, 1986; Hall et al., 2005).

A Appendix A: Supplementary Sample Reconstruction

The sample table below shows the observable route from the broader local file universe to the final estimating sample. It is included to make sample narrowing visible and to show where exclusions enter the paper.

Statistic	Value	RawStatistic
Observations	712,375	rows
Unique Firms	345	unique_firms
Start Year	2003	year_min
End Year	2018	year_max
Mean Xi Real	18.29	mean_xi_real
Mean Cites	8.860	mean_cites
Mean Backward Applicant	6.475	mean_backward_applicant
Mean Backward Examiner	2.132	mean_backward_examiner
Mean Backward Other	0.949	mean_backward_other

Detailed sample reconstruction table.

B Appendix B: Supplementary Descriptive Tables

The descriptive appendix documents the scale, spread, and support of the variables used in the paper. These moments matter because they determine how the estimated coefficients should be read in substantive rather than merely statistical terms.

Variable	N	Mean	SD	P25	Median	P75	Min	Max
Patent quality	712,375	18.29	35.09	3.688	8.255	19.06	0	2718
Forward citations	712,375	8.860	51.01	0.0000	1.000	5.000	0	3222
Log forward citations	712,375	1.135	1.188	0.0000	0.693	1.792	0	8
Applicant backward citations	712,375	6.475	56.62	0.0000	0.0000	1.000	0	2863
Examiner backward citations	712,375	2.132	4.753	0.0000	0.0000	3.000	0	292
Other backward citations	712,375	0.949	9.696	0.0000	0.0000	0.0000	0	1014
CPC group count	711,015	7.350	7.591	3.000	5.000	9.000	1	293
Log patent quality	712,375	2.322	1.051	1.545	2.225	2.999	0	8
Log applicant citations	712,375	0.570	1.085	0.0000	0.0000	0.693	0	8
Log examiner citations	712,375	0.629	0.884	0.0000	0.0000	1.386	0	6
Log other citations	712,375	0.151	0.568	0.0000	0.0000	0.0000	0	7
Log CPC group count	711,015	1.871	0.682	1.386	1.792	2.303	1	6

Descriptive statistics appendix.

C Appendix C: Correlation Matrix

Variable	Patent quality	Forward citations	Log forward citations	Applicant backward citations	Examiner backward citations	Other backward citations	CPC group count	Log patent quality	Log applicant citations	Log examiner citations	Log other citations	Log CPC group count
Patent quality	1.000	0.028	0.010	0.037	0	0.004	0	0.704	0.004	0	0.004	0
Forward citations	0.028	1.000	0.460	0.294	0	0.133	0	0.063	0.090	0	0.095	0
Log forward citations	0.010	0.460	1.000	0.123	0	0.110	0	0.067	-0.065	0	0.192	0
Applicant backward citations	0.037	0.294	0.123	1.000	0	-0.011	0	0.067	0.467	0	-0.030	0
Examiner backward citations	-0.034	-0.002	-0.048	0.086	1	0.017	0	-0.075	0.303	1	0.030	0
Other backward citations	0.004	0.133	0.110	-0.011	0	1.000	0	0.017	-0.051	0	0.603	0
CPC group count	0.065	0.080	0.009	0.109	0	0.009	1	0.078	0.127	0	-0.040	1
Log patent quality	0.704	0.063	0.067	0.067	0	0.017	0	1.000	-0.018	0	0.014	0
Log applicant citations	0.004	0.090	-0.065	0.467	0	-0.051	0	-0.018	1.000	0	-0.139	0
Log examiner citations	-0.064	-0.004	-0.068	0.104	1	0.037	0	-0.128	0.430	1	0.086	0
Log other citations	0.004	0.095	0.192	-0.030	0	0.603	0	0.014	-0.139	0	1.000	0
Log CPC group count	0.077	0.053	-0.039	0.089	0	-0.002	1	0.093	0.152	0	-0.073	1

Correlation matrix for the main numeric variables.

D Appendix D: Yearly Audit

The yearly audit provides a simple check on whether the local panel is driven by a narrow time slice or whether the variables and sample counts evolve over a broad observed window.

Period	Patents	MeanQuality	MeanCites	MeanApplicant	MeanExaminer
2003	33,777	20.10	19.85	0.149	1
2004	32,280	19.40	17.91	0.320	1
2005	34,501	18.93	16.34	0.486	1
2006	34,295	17.26	14.84	0.875	2
2007	37,364	15.33	12.72	1.786	2
2008	40,514	13.03	10.93	2.568	2
2009	34,006	13.72	10.03	3.431	2
2010	36,300	14.36	10.59	5.029	2
2011	41,541	14.73	9.900	6.289	2
2012	51,446	14.73	7.906	7.836	2
2013	52,128	15.59	6.851	8.056	2
2014	52,113	17.01	6.474	10.28	2
2015	56,363	17.78	5.459	10.70	3
2016	58,435	20.57	4.619	11.92	3
2017	60,141	23.63	2.705	10.71	3
2018	57,171	29.66	1.642	9.028	2

Year-by-year sample audit.

E Appendix E: Grouped Outcome Patterns

Grouped summaries do not replace the formal model, but they show the raw outcome structure that the specification is attempting to explain. For that reason they are retained here as descriptive support rather than promoted to the status of a main estimate.

Quantile	N	MeanRegressor	MeanOutcome
Q1	142,511	0.956	0.911
Q2	142,442	1.693	1.144
Q3	142,476	2.228	1.210
Q4	142,471	2.825	1.230
Q5	142,475	3.905	1.182

Quantile summary for the main regressor.

F Appendix F: Full Coefficient Table and Fit Statistics

The full coefficient export is included for completeness so that readers can see the focal terms in the context of the broader specification. Where a fit file is available, it is appended here rather than in the main text to keep the article narrative readable.

term	estimate	std_error	statistic	p_value
log_xi_real	0.0963	0.0167	5.776	¡ .001
log_backward_applicant	0.0779	0.0103	7.554	¡ .001
log_backward_examiner	-0.0491	0.0114	-4.308	¡ .001
log_backward_other	0.1713	0.0134	12.78	¡ .001
log_cpc_group_count	0.2018	0.0175	11.50	¡ .001

Full coefficient dump exported from the local estimation script.

G Appendix G: Coverage Tables

Variable	Missing	MissingShare
Patent quality	0	0.00%
Forward citations	0	0.00%
Log forward citations	0	0.00%
Applicant backward citations	0	0.00%
Examiner backward citations	0	0.00%
Other backward citations	0	0.00%
CPC group count	1,360	0.19%
Log patent quality	0	0.00%
Log applicant citations	0	0.00%
Log examiner citations	0	0.00%
Log other citations	0	0.00%
Log CPC group count	1,360	0.19%

Missingness audit for the variables summarized in the manuscript.

Entity	Observations	FirstPeriod	LastPeriod
12490	99,013	2003	2018
10107	36,662	2003	2018
59328	33,177	2003	2018
12060	26,285	2003	2018
77178	25,638	2003	2018
90319	24,004	2003	2018
27828	22,181	2003	2018
14593	21,058	2003	2018
53613	18,069	2003	2018
66093	15,747	2003	2018

Entity	Observations	FirstPeriod	LastPeriod
25785	15,307	2003	2018
84788	15,195	2003	2018
19561	13,600	2003	2018
15579	12,713	2003	2018
12369	12,654	2003	2018
76076	12,619	2003	2018
10145	11,021	2003	2018
17830	10,275	2003	2018
10104	9,264	2003	2018
61241	8,473	2003	2018

Top entities by local observation count.

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